

Joyce A. Chew

Ph.D. candidate in Mathematics

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Research Interests

Learning on manifolds, low-rank matrix and tensor approximations, bias in machine learning, applications of interpretable machine learning methods

Education

- 2025 **Ph.D in Mathematics**, *University of California, Los Angeles*
(expected) Advisor: Deanna Needell
Thesis title (tentative): Topics in Learning on Manifolds
- 2023 **C.Phil. in Mathematics**, *University of California, Los Angeles*
- 2022 **M.A. in Mathematics**, *University of California, Los Angeles*
- 2020 **B.S. in Mathematics, Honors**, *Calvin University*, Grand Rapids, MI
- 2020 **B.A. in Chemistry**, *Calvin University*, Grand Rapids, MI

Honors and Awards

- 2024 SIAM Student Travel Award (LA24)
- 2022 UCLA Raymond Redheffer Prize (awarded for excellence in teaching calculus)
- 2020 NSF Graduate Research Fellowship
- 2019 National Center for Women and Information Technology Collegiate Award Finalist
- 2018 Goldwater Scholarship Honorable Mention
- 2016 Calvin University Student Research Fellowship

Publications

* denotes undergraduate author.

Journal Papers

- [A1] **J. A. Chew**, M. Hirn, S. Krishnaswamy, D. Needell, M. Perlmutter, H. Steach, S. Viswanath, and H.-T. Wu. Geometric scattering on measure spaces. *Applied and Computational Harmonic Analysis*, page 101635, 2024. doi:10.1016/j.acha.2024.101635.
- [A2] K. Cheng*, S. Inzer*, A. Leung*, X. Shen*, M. Perlmutter, M. Lindstrom, **J. A. Chew**, T. Presner, and D. Needell. Multi-scale hybridized topic modeling: A pipeline for analyzing unstructured text datasets via topic modeling.

SIAM Undergraduate Research Online, 16, April 2023. doi:10.1137/22S1536832.

- [A3] N. P. Kazmierczak, **J. A. Chew**, and D. A. Vander Griend. Bootstrap methods for quantifying the uncertainty of binding constants in the hard modeling of spectrophotometric titration data. *Analytica Chimica Acta*, 1227:339834, September 2022. doi:10.1016/j.aca.2022.339834.
- [A4] N. P. Kazmierczak, **J. A. Chew**, and D. A. Vander Griend. A reliable algorithm for calculating stoichiometry parameters in the hard modeling of spectrophotometric titration data. *Journal of Chemometrics*, 36(6):e3409, May 2022. doi:10.1002/cem.3409.
- [A5] P. Li*, C. Tseng*, Y. Zheng*, **J. A. Chew**, L. Huang, B. Jarman, and D. Needell. Guided semi-supervised non-negative matrix factorization. *Algorithms*, 15(5):136, April 2022. doi:10.3390/a15050136.
- [A6] N. P. Kazmierczak, **J. A. Chew**, A. R. Michmerhuizen, S. E. Kim, Z. D. Drees, A. Rylaarsdam, T. Thong, L. Van Laar, and D. A. Vander Griend. Sensitivity limits for determining 1: 1 binding constants from spectrophotometric titrations via global analysis. *Journal of Chemometrics*, 33(5):e3119, 2019. doi:10.1002/cem.3119.
[Refereed Conference/Workshop Papers](#)
- [B1] D. R. Johnson, **J. A. Chew**, S. Viswanath, E. De Brouwer, D. Needell, S. Krishnaswamy, and M. Perlmutter. Convergence of manifold filter-combine networks. 2024. doi:10.48550/arXiv.2410.14639. Accepted to NeurIPS Workshop on Symmetry and Geometry in Neural Representations (Extended Abstract Track) 2024.
- [B2] A. Venkat, **J. A. Chew**, F. C. Rodriguez, C. J. Tape, M. Perlmutter, and S. Krishnaswamy. Directed scattering for knowledge graph-based cellular signaling analysis. 2023. doi:10.48550/arXiv.2309.07813. Accepted to ICASSP 2024.
- [B3] **J. A. Chew**, D. Needell, and M. Perlmutter. A convergence rate for manifold neural networks. In *2023 International Conference on Sampling Theory and Applications (SampTA)*, pages 1–5. IEEE, 2023. doi:10.1109/SampTA59647.2023.10301407.
- [B4] **J. A. Chew**, H. Steach, S. Viswanath, H.-T. Wu, M. Hirn, D. Needell, M. D. Vesely, S. Krishnaswamy, and M. Perlmutter. The manifold scattering transform for high-dimensional point cloud data. In *Topological, Algebraic and Geometric Learning Workshops 2022*, volume 196, pages 67–78. PMLR, 2022. URL <https://proceedings.mlr.press/v196/chew22a.html>.
[Preprints](#)
- [C1] **J. A. Chew**, W. Diepeveen, and D. Needell. Curvature corrected nonnegative manifold data factorization. *arXiv preprint*, 2025. doi:10.48550/arXiv.2502.15124.
- [C2] D. R. Johnson, **J. A. Chew**, E. De Brouwer, S. Krishnaswamy, D. Needell, and M. Perlmutter. Manifold filter-combine networks. *arXiv preprint*, 2023. doi:10.48550/arXiv.2307.04056.

- [C3] E. George, **J. A. Chew**, and D. Needell. Detecting and mitigating indirect stereotypes in word embeddings. *arXiv preprint*, 2023. doi:10.48550/arXiv.2305.14574.
- [C4] W. Diepeveen, **J. A. Chew**, and D. Needell. Curvature corrected tangent space-based approximation of manifold-valued data. *arXiv preprint*, 2023. doi:10.48550/arXiv.2306.00507.

Teaching and Research Mentorship

- 2023–2024 **Lead Mentor, UCLA/Los Angeles Pierce College Collaborative Undergraduate Research Experience**
 Funded by NSF DUE 2150195 (abstract).
 PIs: Edouard Tchertchian, Michael Hill, Jianzhong Su, and Padmanabhan Seshaiyer.
 Project: Evaluating Randomized Kaczmarz Methods
- 2022 **REU Mentor, UCLA CAM REU**
 Co-mentors: Michael Perlmuter (lead), Michael Lindstrom, Todd Presner (co-PI), Deanna Needell (co-PI)
 Project: AI for Holocaust Studies (paper: [A2])
- 2021–2025 **Teaching Assistant, UCLA Department of Mathematics**
 Courses: Calculus I, Calculus II, Applied Numerical Methods, Experience of Data Science
- 2021 **REU Mentor, UCLA CAM REU**
 Co-mentors: Longxiu Huang (lead), Benjamin Jarman, Deanna Needell (PI)
 Project: Data Science for Innocence (paper: [A5])

Other Mentoring

- 2021–2023 **UCLA Women in Mathematics Undergraduate Mentoring**
- 2021–2022 **UCLA Mathematics Directed Reading Program**
 Topics: Convex optimization, Kaczmarz methods, numerical linear algebra

Talks

- 2024 **Curvature corrected semi-nonnegative matrix factorization of manifold-valued data**, *SIAM LA24, SIAM IS24, MAA Mathfest 2024*
- 2023 **A Convergence Rate for Manifold Neural Networks**, *SampTA 2023*
- 2023 **Uncovering Structure in High-Dimensional Data**, *Calvin University Mathematics and Statistics Colloquium*
- 2022 **Geometric Scattering on Non-Euclidean Data**, *SIAM MDS22*
- 2022 **NSF GRFP Workshop**, *UCLA Department of Mathematics*
- 2019 **Slinkies, Gorges, and Ice Cream: What I Did on My Summer Vacation**, *Calvin University Mathematics and Statistics Colloquium*
- 2019 **Tension induced instabilities of twisted springs**, *Cornell University Undergraduate Research Forum*
- 2015 **Cake and what I learned from cutting it**, *TEDxValencia High School*

Professional Memberships

SIAM, AWM

Programming Languages

Python, C, C++, MATLAB, Julia, R